

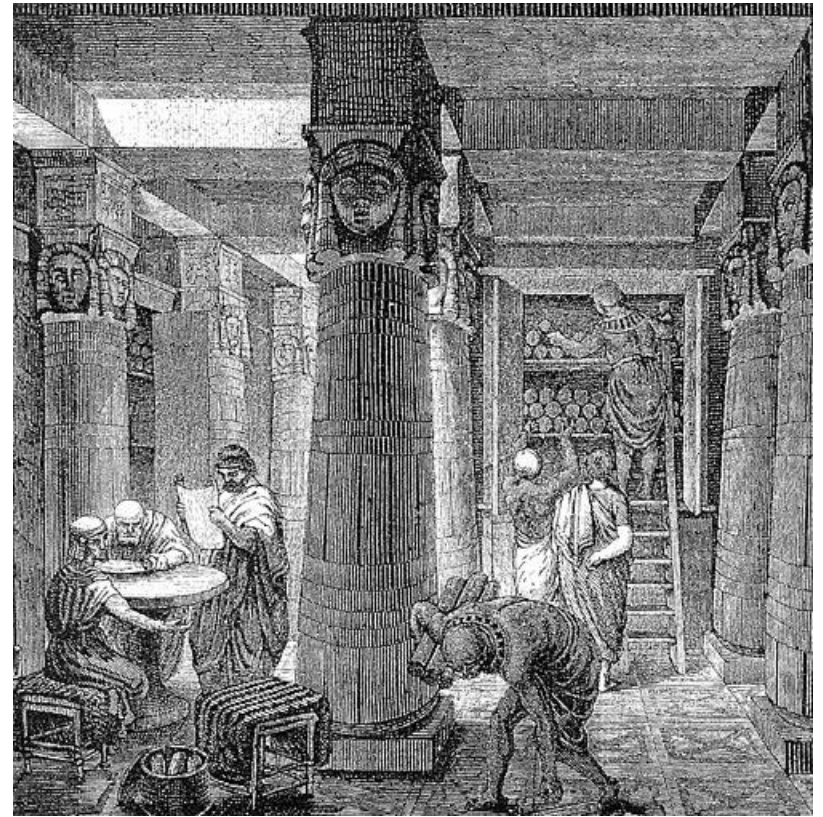


A History of the Information (Data) Quality Discipline

Elizabeth Pierce

Information in Ancient Times

- Library and information science deal with the organization, access, collection, and management of information.
- The world's oldest known library was founded sometime in the 7th century B.C. Located in Nineveh in modern-day Iraq, the site included a trove of some 30,000 cuneiform tablets organized according to subject matter.



Over time as our information expanded, issues over its quality emerged.

- Economics: How to address data quality issues in the federal statistics system.
- Information & Library Sciences: How to implement quality control in databases & catalogs.
- Survey Research: How to address quality issues surrounding the editing and imputation of survey data.



Data Quality became a topic of research across multiple disciplines

■ Criminal Justice

- How to effectively manage criminal records.
- How to detect and prosecute cyber (ID) crime.

■ Accounting/Auditing

- How to better prevent, detect, and correct monetary errors.

■ Computer Science

- How to design and implement database systems to improve the security and integrity of the data.

■ Management/MIS

- How to address poor quality business data.
- How to move data faster and more efficiently.

Information Quality Emerges - A Recap of Important, Early Events

- **Early 1990's:** Larry English founds Information Impact International, a company whose mission is to provide quality education and consulting in information management.
- **1992:** MIT officially established Total Data Quality Management (TDQM) program with Honeywell and Fujitsu as the founding sponsors, followed by D&B, Firstlogic, Peacehealth Corporation, Georgia Division of Public Health, US Naval Supply Systems Command, and many others during the 1990's.
- **1996:** First International Conference on Information Quality (ICIQ) held at MIT, Cambridge, Mass.

State of IQ Research circa 1995

- Management Responsibilities
 - Development of a corporate data quality policy
 - Establishment of a data quality system
- Operation and Assurance Costs
 - Operating costs include prevention, appraisal, and failure costs
 - Assurance costs relate to the demonstration and proof of quality as required by customers and management
 - Area in which this work occurs (i.e. Information Systems, Databases, and Accounting)

State of IQ Research circa 1995

■ Research and Development

- Definition of the dimensions of data quality and measurement of their values
- Analysis and design of the quality aspects of data products
- Design of data manufacturing systems that incorporate data quality aspects

■ Production

- Quality requirements in the procurement of raw data, components, and assemblies needed for the production of data products
- Quality verification of raw data, work-in-progress, and final data products
- Identification of non-conforming data items and specifications of corrective actions

State of IQ Research circa 1995

■ Distribution

- Storage, identification, packaging, installation, delivery, and after-sales servicing of data products
- Quality documentation and records for data products

■ Personnel Management

- Employee awareness of issues related to data quality
- Motivation of employees to produce high-quality data products
- Measurement of employee's data quality achievement

■ Legal Function

- Data product safety and liability

Information Quality Field – A Recap of Important, Early Events

- **1997:** Thomas Redman founds Navesink Consulting Group. Among its many IQ services, Navesink Consulting Group creates the Data Quality College, a series of seminars that focus on data quality issues.
- **1999:** O. Khalil, D. Strong, B. Kahn and L. Pipino author the paper entitled “Teaching Information Quality in Information Systems Undergraduate Education” in Informing Science.
- **2001:** Craig Fisher offers the first undergraduate course on IQ at Marist College, Poughkeepsie, NY.
- **2001:** Europe’s launches its own annual Data Management & IQ Conferences in London, UK.

Information Quality Field – A Recap of Important, Early Events

- **2002:** The MIT-IQ program offers its first three day workshop entitled MIT-IQ for Executives.
- **2002:** WooYoung Chung, Craig Fisher, and Richard Wang present “Redefining the Scope and Focus of Information-Quality Work: A General Systems Perspective” at ICIQ classifying IQ skills into three categories: Technical, Adaptive, and Interpretive.
- **2003:** The MIT-IQ program offers its first week-long workshop entitled “IQ-1: Principles & Foundations” .
- **2004:** International Association for Information and Data Quality, a not-for profit, vendor-neutral professional society is established.

Information Quality Field – A Recap of Important, Early Events

- **2004:** CAiSE sponsors the Workshop on Data and Information Quality in Riga, Latvia. A second DIQ workshop is held by CAiSE in 2005.
- **2004:** SIGMOD sponsors the International Workshop on Information Quality in Information Systems (IQIS) in Paris, France. IQIS 2004 is followed by IQIS 2004 and IQIS 2005.
- **2005:** Diane Strong, Craig Fisher, David Feinstein, and Herbert Longenecker publish their article “Teaching, Learning, and Curriculum Development to Support Managing Information as a Product” in the AMIS Monograph on Information Quality. This article explores various strategies for incorporating IQ into an IS or Business UG Education.

Information Quality Field – A Recap of Important, Early Events

- **2005:** Acxiom agrees to provide multi-year financial support for a Masters in Information Quality, UALR agrees to house the program in the Department of Information Science, and the MITIQ Program agrees to guide curriculum development.
- **2005:** Deborah Henderson, Pat Cupoli, Jeff Hoffer, Anne Marie Smith, Eva Smith, (DAMA International - “ Model Curriculum Framework for Post Secondary Education Program in Data Management”) ISECON 2005 Meritorious Writing Award. Recommended the addition to IS2002 standard: Data Warehousing, Metadata Management and Data Quality.
- **2006:** University of Arkansas at Little Rock launches the first Master of Science in Information Quality with 25 students.

Information Quality Field – Recap of Important, Early Events

- **2006:** At Northeastern University, Dr. Yang Lee teaches the Honors Seminar: “Information Quality: Technology and Philosophy.
- **2007:** The International Journal of Information Quality published by InderScience releases its first issue.
- **2007:** University of South Australia, School of Computer and Information Science announces plans to launch a Master of Science in Information Quality.
- **2007:** University of Westminster in the UK announces plans to launch a Master of Science in Information Quality.

Information Quality Field – A Recap of Important, Early Events

- **2007:** The ACM Journal of Data and Information Quality begins accepting submissions.
- **2007:** The University of Arkansas at Little Rock posts its Fall 2007 schedule with all courses required for the MS in Information Science offered online as part of its distance education program.
- **2007:** Yang Lee offers the first Graduate course at Northeastern University on Information Quality based on the book *Journey to Data Quality* (2006, MIT Press): “Information Quality for Global Managers”.
- **2009:** ICIQ goes global (Postdam, Germany (2009), Little Rock (2010 and 2013), Adelaide, Australia (2011), Paris, France (2012), Xi'an, China (2014).

Areas of IQ Research circa 2007

- Foundational Information Quality Research Topics
- Information Curation
- Database related technical solutions for IQ
- Entity Resolution
- IQ in the context of Computer Science and Information Technology
- IQ in the Enterprise Context

2007: Foundational IQ Research Topics

- Continued work in the IQ Research Areas identified by Wang and others in their 1995 Analysis of IQ Research Areas
- IQ Practices: Case Studies and Experience Reports
- Environment Specific Information Quality
 - Scientific Data Management, Healthcare, Government, Education, etc.
- IQ Education and Curriculum Development

2007: Information (Digital) Curation

- Example Problem: NASA's loss of the original tapes of Apollo 11 Moon Landing
- Research Opportunities
 - Fragility of Digital Media
 - Technological Obsolescence
 - Lack of good practices (i.e. adequate documentation)
- Interested in learning more?
 - International Journal of Digital Curation
<http://www.ijdc.net/ijdc/issue/current>
 - Digital Curation Centre and its research agenda
<http://www.dcc.ac.uk/>

2007: Database related technical solutions for Information Quality

- Example Problem: Improving the IQ of Databases, the Web, and e-business
- Research Opportunities
 - New types of database systems that manage data and uncertainty associated with the data
 - Data lineage and provenance
 - Data Cleansing and Reconciliation
 - Enterprise Architecture Deployment
 - IQ in Data Warehouses and Data Mining
- Interested in learning more?
 - Trio at <http://infolab.stanford.edu/trio/>

2007: Entity Resolution

- Example Problem: How to connect two or more records when there is no common identifying key
- Research Opportunities
 - Entity Resolution Algorithms that address Matching and Merging functions, Performance, Distribution, Secondary Storage, Numerical confidences, and the handling of Negative information as well as positive
- Interested in learning more?
 - Stanford Entity Resolution Framework
<http://infolab.stanford.edu/serf/>
 - The Laboratory for Advanced Research in Entity Resolution and Information Quality (ERIQ)
<https://ualr.edu/eriq>

2007: IQ in the context of Computer Science and Information Technology

- Example Problem: How can we incorporate IQ principles into our data production processes?
- Research Opportunities
 - New ways of understanding, modeling, improving, and incorporating IQ (i.e. better Tools, Metrics, Measures, Models, and Methodologies)
 - Technical solutions of information systems (i.e. better information product implementation, delivery, and management)
 - Technical layers of networks and communications
 - Data Privacy and Protection mechanisms
- Interested in Learning More?
 - IQ at MIT <http://mitiq.mit.edu>

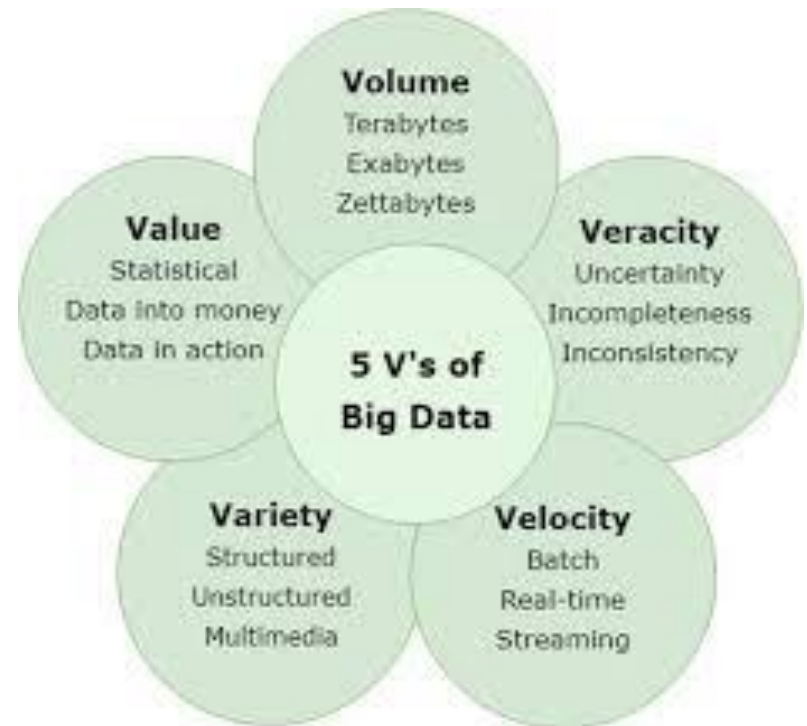
2007: IQ in the Enterprise Context

- Example Problem: Information for many organizations is their most poorly governed asset
- Research Opportunities
 - Impact and role of IQ on business, work process, policies and standards, and strategy
 - Impact and role of IQ on a firm's overall operational or economic performance, cost and benefits, IT mgmt, HR mgmt
 - Impact of IQ on knowledge mgmt, customer mgmt, supply-chain mgmt, extended-enterprise mgmt, and global mgmt
 - Impact and role of IQ in groups, organizations, and society
- Interested in Learning More?
 - Data Governance Institute
<http://www.datagovernance.com/>

The IQ Field Continues to Grow

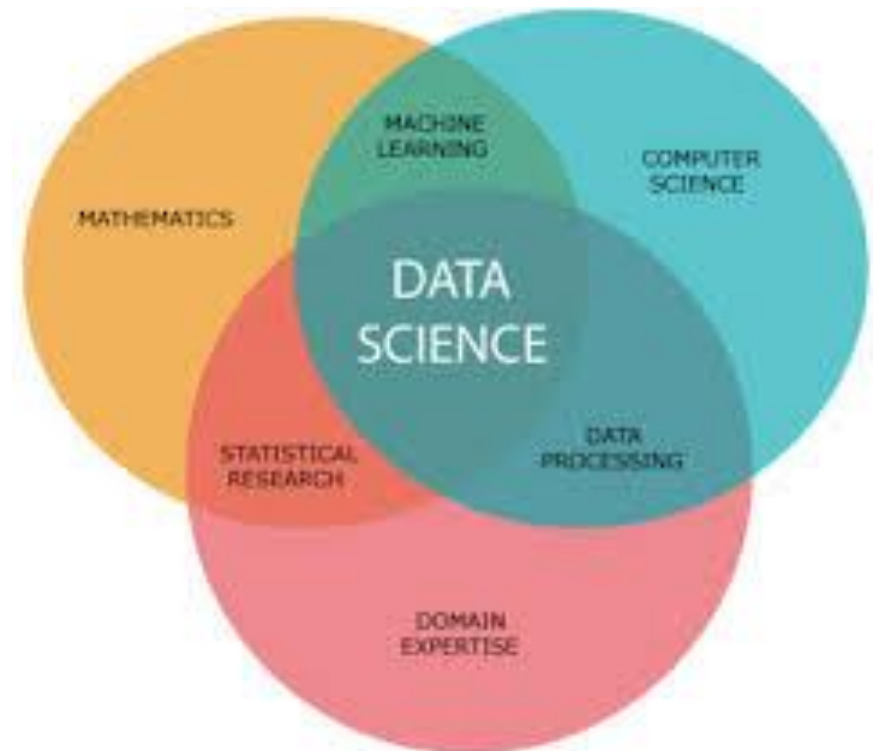
■ Big Data Issues

- How to Assess DQ
- How to Prevent, Detect, Correct Data Issues
- How to Manage Data Lakes and Streams
- Ethics/Security/Privacy/Crossing the Creepy line



IQ's role in Data Science

- Data Analytics needs good data.
- Strategies/Best Practices for Ensuring High Quality Data
- Tools and Technologies for Maintaining Data Quality



Information as an Asset (Quality)

- First CDO was appointed at Capital One in the early 2000's. Today about 25% of organizations have a CDO.
- The first generation was mostly about defense-driven, risk-mitigating endeavors.
- The second (current) generation, is about bringing business value out of data, and the emphasis is on analytics and data science.



Managing New Types of Data

- Web and social media data such as Facebook, Twitter, LinkedIn, and blogs (also includes clickstream data).
- Machine to machine data such as IoT sensors, meters, or RFID that report events such as speed, temperature, pressure, flow, moisture content, salinity, and geolocation.
- Big transaction data that includes healthcare claims, telecommunications call detail records, and utility billing records.
- Biometrics that include anatomical data (e.g., fingerprints, iris, retina, face, hand outline, ear shape, voice pattern, DNA, body odor) and behavioral data (e.g., handwriting and keystroke analysis).
- Human generated data such as call center agents' notes, voice recordings, email, paper documents, surveys, and electronic medical records.

New Data Management and Integration Concepts

- **Data catalog:** An organized inventory of an organization's data assets that allows users to search, discover, and access data. Example: <https://www.alation.com/product/data-catalog/>
- **Data mesh:** A decentralized architecture that aims to build business-focused data products without specifying technology. It's designed to reduce friction in data access and promote collaboration, with a user-centric approach to data management. Example: <https://aws.amazon.com/what-is/data-mesh/>
- **Data fabric:** An architecture and infrastructure that integrates data from various sources and systems into a unified layer, enabling seamless data interoperability and real-time analytics. It takes a more automated, product-and-tech-centric approach to data management, with a focus on centralization and unified data access. Example: <https://www.ibm.com/topics/data-fabric>

IQ and Artificial Intelligence

- AI systems rely on large amounts of data to learn and perform tasks.
- Poor data quality can lead to biased, unreliable, or unethical AI outcomes.



IQ Research Collaboration

Opportunities Continue to Expand

- Role of big data in medicine is one where we can build better health profiles and better predictive models around individual patients so that we can better diagnose and treat diseases.
- Biologists are finding themselves unable to extract full value from the large amounts of data becoming available.
- Social Media Mining and Analytics are facing DQ issues.
- Growing opportunities for historians as old books are scanned and organized.
- The legal system generates a huge and ever-increasing amount of data.
- Imagine classrooms outfitted with cameras, microphones, and fit bits that run constantly, capturing each child's every facial expression, fidget, and social interaction, every day, all year long.

New Data Quality Careers

- AI, ML, Data Science Jobs that related to Data Quality - <https://ai-jobs.net/list/data-quality-related-jobs/>
- Data Quality Analyst Jobs - <https://www.indeed.com/q-data-quality-analyst-jobs.html?vjk=a90e7b27a7657bb1>
- New Executive roles (CIO/CISO/CTO/CDO) - <https://www.linkedin.com/pulse/cio-ciso-cto-whats-difference-preon-evans/>

Any Questions?
